

Chemineers

Computing the limit of ΔT_{LMTD}

→ Get the value of ΔT_{LMTD} when heat capacity rates are equal ($\dot{C}_c = \dot{C}_h$)

→ In a counter-flow heat exchanger

a) Hot fluid enters at $T_{h,in}$ and leaves at $T_{h,out}$.

b) Cold fluid enters at $T_{c,in}$ (opposite end) and leaves at $T_{c,out}$.

Temperature difference at the two ends are defined as:

$$\Delta T_1 = T_{h,in} - T_{c,out}$$

$$\Delta T_2 = T_{h,out} - T_{c,in}$$

→ The total heat transfer rate $\dot{Q}$ is given by:

$$\dot{Q} = \dot{C}_h (T_{h,in} - T_{h,out}) = \dot{C}_c (T_{c,out} - T_{c,in})$$

$$\Rightarrow T_{h,in} - T_{h,out} = T_{c,out} - T_{c,in} \quad [\because \dot{C}_h = \dot{C}_c]$$

$$\Rightarrow T_{h,in} - T_{c,out} = T_{h,out} - T_{c,in}$$

$$\Rightarrow \boxed{\Delta T_1 = \Delta T_2}$$

By definition,

$$\Delta T_{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1) - \ln(\Delta T_2)}$$

as, $\Delta T_1 = \Delta T_2 \rightarrow$ it becomes $\frac{0}{0}$ form. We have to calculate:

$$\lim_{\Delta T_2 \rightarrow \Delta T_1} \Delta T_{LMTD} = \lim_{x \rightarrow c} \left(\frac{c-x}{\ln c - \ln x} \right)$$

(Applying L'Hospital)

$$= \lim_{x \rightarrow c} \frac{\frac{d}{dx}(c-x)}{\frac{d}{dx}(\ln c - \ln x)} = \lim_{x \rightarrow c} \frac{-1}{-\frac{1}{x}} = c$$

$$= \Delta T_1 = \Delta T_2 = \Delta T$$

Let, $\Delta T_2 = x$
 $\Delta T_1 = c$
 (a constant)

$\therefore$ In a balanced counter-flow heat exchanger, the LMTD is simply the constant temperature difference between the two fluids across the entire length of the exchanger.

Why Exponential Behaviour Fails:-

The general differential equation for the temperature difference $\theta = T_h - T_c$ along the area A is derived from energy balance on a differential element dA .
 for parallel flow heat exchanger (gradient adds up),

$$d\theta = -U dA \left(\frac{1}{\dot{C}_h} + \frac{1}{\dot{C}_c} \right) \theta$$

But for counter-flow heat exchanger (fluids move in opposite direction),

$$d\theta = -U dA \left(\frac{1}{\dot{C}_h} - \frac{1}{\dot{C}_c} \right) \theta$$

Here, $\dot{C}_h = \dot{C}_c$, so, $d\theta = 0$

$$\theta(x) = T_h(x) - T_c(x) = \text{constant}$$